RAJALAKSHMI ENGINEERING COLLEGE

Ancilin Rose R X 251801021-2025-AIDS A2

Started on	Thursday, 27 August 2026, 11:09 AM
State	Finished
Completed on	Thursday, 27 August 2026, 11:16 AM
Time taken	7 mins 46 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
int main()
{
    int n, a[100], x;
    int i, low, high, mid, found = 0;
    scanf("%d", &n);
    for(i = 0; i < n; i++)
        scanf("%d", &a[i]);
    scanf("%d", &x);
    for(i = 0; i < n; i++)
    {
        int need = x - a[i];
        low = i + 1;
        high = n - 1;
        while(low <= high)
        {
            mid = (low + high) / 2;
            if(a[mid] == need)
            {
                printf("%d\n", a[i]);
                printf("%d", a[mid]);
                found = 1;
                break;
            }
            else if(a[mid] < need)
                low = mid + 1;
            else
                high = mid - 1;
        }
        if(found == 1)
            break;
    }
    if(found == 0)
        printf("No");
    return 0;
}
```

	Input	Expected	Got	
✔	4 2 4 8 10 14	4 10	4 10	✔

	Input	Expected	Got	
✔	5 2 4 6 8 10 100	No	No	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course